# Global Dynamic Path Planning Based on Fusion of Improved A* Algorithm and Morphin Algorithm

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**Abstract:** To meet the requirements of global optimal and real-time obstacle avoidance in mobile robot path planning, A novel method based on the fusion of improved A* algorithm and Morphin search tree algorithm is proposed. Firstly, A more appropriate heuristic function is designed for A* algorithm; Then a key node culling scheme is introduced into the traditional A* algorithm to remove the redundant nodes; Finally, the Morphin algorithm is integrated to perform real-time dynamic path planning on the global path based on the local information collected by the mobile robot sensor. On the basis of ensuring the better global path, the smoothness and local obstacle avoidance ability of path planning are improved. MATLAB simulation tests show that the mixed algorithm with A* and Morphin search tree algorithm can effectively shorten the path length and improve the efficiency to reach the target in the autonomous mobile robot path planning, efficiently complete path planning.

**Key Words:** A* algorithm; Morphin searching tree; path planning; dynamic obstacle avoidance

## 1 INTRODUCTION

Path planning problem is a research hotspot in the field of intelligent robot. Mobile robot path planning refers to planning the best or sub-optimal safe collision-free path from the start point to the end point in a complex environment. The path planning of mobile robot is divided into two types: global path planning method and local path planning method. At present, the path planning algorithms mainly focus on the following methods: Traditional methods mainly include D* algorithm, artificial field method, RRT (Rapidly-exploring random tree) algorithm, etc [2-4]. Intelligent algorithms mainly include neural network algorithms [5], particle swarm optimization algorithms, and ant colony algorithms and so on. These algorithms have their own advantages and disadvantages. For example, the artificial potential field method is simple to implement, but it is easy to fall into a local minimum; The particle swarm optimization algorithm [6] has the characteristics of fast convergence speed and few setting parameters, but the optimization efficiency is poor. Ant colony algorithm [7] has strong generality and robustness, but it is computational overhead is large. Domestic and foreign scholars have proposed improved methods for the shortcomings of different algorithms. Literature [6], the paper presents a hybrid path planning with Morphin algorithm is proposed, and the QPSO adaptive local search strategy is improved. Crossover operation is introduced to improve performance and avoid local optimum. Among them, the Morphin search tree [8] algorithm has small computation and good practicability. It is suitable for handling path planning problems with dynamic obstacles. In the literature [9], the A* algorithm is optimized by using the double-layer raster map method. The high-rise A* algorithm determines the next large grid local map, and the low-level A* algorithm performs path planning on the merged local map. A* Algorithm is a fast, widely used path planning algorithm. For the traditional A* algorithm, there are many nodes with large corners, and it is impossible to avoid obstacles dynamically. In order to solve the above problems， this paper proposes a fusion algorithm that combines the improved A* algorithm with the Morphin algorithm. The algorithm mainly uses the improved A* algorithm to plan the global path, and uses the Morphin search tree algorithm to modify the local path. When encountering a dynamic obstacle, the mobile robot can perform local path planning.After bypassing the obstacle, return to the global path to achieve dynamic obstacle avoidance.

## 2 GLOBAL PATH PLANNING BASED ON IMPROVED A* ALGORITHM

### 2.1 A* Algorithm Description

The A* algorithm is a classic heuristic search algorithm. It uses the heuristic function to select the lowest cost of searching for the next node from the current node. It repeats the loop until the search for the target point is terminated. The evaluation function in the heuristic search process is expressed as：

$$f(n) = g(n) + h(n) \tag{1}$$

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Formula (1)： *f (n)* represents the lowest cost estimate from the starting point through node *N* to the target point. It consists of two parts: the first part *g (n)* is the minimum actual cost between the starting point and the *N* node; the second part *h (n)* is the estimated cost of the current node n point to the target point.

A* algorithm needs to constantly update two lists during the search process. One is an open list and the other is a closed list. The open list only saves subsequent nodes of the current node, and does not save subsequent nodes of other extended nodes. The A* algorithm selects the node with the lowest cost estimate as the parent node in the open list. The closed list is the parent node that passes through and obstacle nodes.

### 2.2 Improved A * Algorithm

The traditional A* algorithm [9] has many disadvantages, such as many search nodes, many polylines and large turning angles. In order to solve the above problems, this paper mainly improves the A * algorithm [10-11] in two aspects. The first point is to improve the selection of key nodes and delete redundant nodes. The second node of the planned path begins to select nodes. Determines that the current node is on the same line as the previous node and its parent. If the previous node is deleted as a redundant point on the same row, the path is updated. The second improvement is the smoothing of the planning path [12]. In the literature [12], the traditional A* algorithm produces a large steering angle and the mobile robot is difficult to rotate. The creation function is used to determine whether there are obstacles between the connection points. If there are no obstacles, the intermediate redundant nodes can be deleted after multiple judgments. In this paper, the method of moving average filtering is used to smooth the path. This method is simple and effective. The moving average filtering smoothing processing formula is as follows:

$$N(1) = x(1) \tag{2}$$

$$N(2) = (x(1)+x(2)+x(3))/3 \tag{3}$$

$$N(3) = (x(1)+x(2)+x(3)+x(4)+x(5))/5 \tag{4}$$

$$N(4) = (x(2)+x(3)+x(4)+x(5)+x(6))/5 \tag{5}$$

*N( )* is new data obtained using a moving average filter. And *x( )* is an element in the data waiting to be processed. The default span for the moving average is 5.

Based on statistical laws, continuous sampled data is treated as a queue of fixed length *N*. After a new measurement, the first data of the above queue is removed. The remaining *N-1* data are advanced in turn, and the new sample data is inserted as the tail of the new queue. Then perform an arithmetic operation on the new queue and use the result as the result of this measurement. Tracking time and path length are reduced after smoothing. After the planned path is smoothed, the path-finding time and path length are significantly improved. The traditional A* algorithm can better perform global path planning after the above two improvements.

## 3 MORPHIN ALGORITHM AND LOCAL PATH PLANNING

### 3.1. Morphin Algorithm Description

Morphin algorithm is a local path planning algorithm based on ground analysis proposed by CMU according to Ranger [13] algorithm. It is also a local obstacle avoidance algorithm based on a prior grid environmental map for feasibility statistical analysis [14-15]. The core idea is to detect the environmental information in real time according to the robot's own sensor. And generate a set of travelable path sets according to different steering angles. Then analyze the steering angle of the robot and the passage rate and safety of the front path. Select the most appropriate path based on the current state. The algorithm not only has less computational complexity and faster efficiency, but also better handles the uncertainty of environmental modeling. It can also be combined with the global path planning algorithm to make a common decision on the robot's motion behavior. The prediction path of the Morphin algorithm is shown in Figure 1.The circle is the robot, the diamond is the target, and the black block is the obstacle.

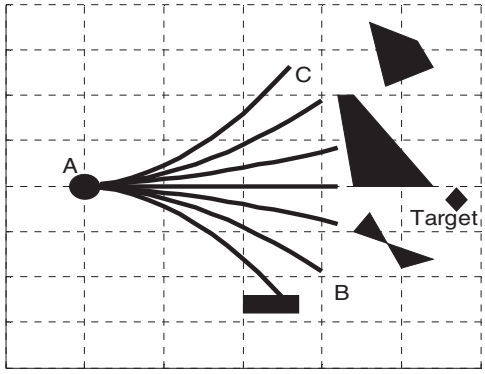



Fig.1 Morphin Algorithm path selection

It can be seen from Figure 1, that robot chooses the path with the least chance of collision in order to avoid obstacles. Among the 7 arcs, AB and AC are alternative paths with no collision. The AC path to the target point can result in a complicated path. Therefore the AB path is more reasonable. Multiple alternative path plans generated by Morphin search tree algorithm. We decided to use the best of the evaluation function. And the evaluation function is:

$$f = \begin{cases} \infty, & Obstructions\ \ on\ \ the\ \ arc \\ \gamma_1 L + \gamma_2 M + \gamma_3 \Delta d + \gamma_4 N, & No\ \ \ Obstacle \end{cases} \tag{6}$$

Where *L* and *M* are the length of the arc and the number of inflection points; $\Delta d$ is the average distance from each grid point of the arc to the sub-target point; *N* is the number of intersections between the end point of the arc and the sub-target point and the obstacle grid； $\gamma_1$， $\gamma_2$， $\gamma_3$ and $\gamma_4$ are the weight of each parameter. *f* infinity when there is an obstacle in the arc. When *f* is the minimum value, the corresponding arc is the optimal path for local planning.



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### 3.2. Local Planning and Dynamic Path Planning Process

The flow of this algorithm is shown in figure 2. The robot firstly performs global planning based on the improved A* algorithm. It uses its own real-time sensor detects the ambient information, and perceive the existence of an unknown obstacle. Predict the trajectory and speed of the obstacle and determine if the robot will collide with the obstacle. If the car is about to collide with an unknown obstacle, use the Morphin algorithm for local planning to bypass the obstacle.

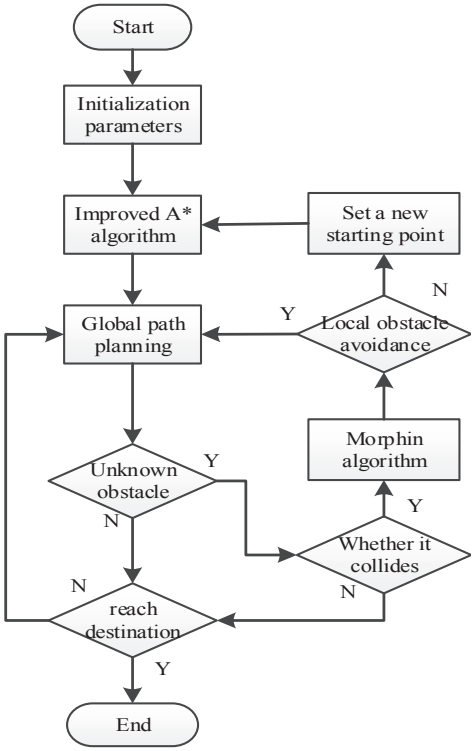



Fig2: Fusion algorithm flow chart

The Morphin algorithm in the fusion algorithm is based on the scrolling window principle. It uses the idea of time scrolling window. Set the robot-centric work area based on the grid environment information of the robot's work. Real-time sensors collect ambient information of 3 grid distances at regular intervals. The robot predicts whether the next movement will encounter an unknown obstacle. There will be collisions, If a collision is about to occur, the robot adjusts the trajectory and speed of the running parameters in time. The local planning algorithm steps are as follows:

(1) The robot updates the gird map information at intervals according to the detection of obstacle information, and saves the current node. There are no obstacles in front and no collision. Go to step (6).

(2) Determine multiple sub-target points if a collision is imminent.

(3) The current node is the starting point and draws a predicted arc pointing to the sub-target point. As shown in Figure 1. The arc is represented in a nearby gird.

(4) According to the evaluation function *f*, the arc with the smallest value is selected from the candidate path become the optimal path. Go to Steps (7); Perform step (5) when several prediction paths cannot avoid obstacles.

(5) The robot rotates at a certain angle or waits for a while to plan again. If the failure stops the local path planning. Re-plan globally using the A* algorithm and follow the new path to reach the target point.

(6) If the unknown obstacle does not affect the planned global path, the robot will travel as planned.

(7) The robot successfully bypasses the obstacle and returns to the global path. It continues to travel to the target point.

## 4 SIMULATION RESULTS AND ANALYSIS

In order to verify the effectiveness of the global path planning and local path planning proposed in this paper. A series of simulation experiments are carried out. Firstly, the effect of traditional A* algorithm and improved A* algorithm in global path planning is compared and analyzed. Then add some dynamic obstacles. Using Morphin algorithm for local path planning in order to correct the global path. The simulation experiment was carried out on the experimental platform of MATLAB R2014a. The computer is configured with CPU 2.5Ghz and 4G memory. Comparison experiment of multiple groups of traditional A* and improved A* under grid map model 20x20 (grid spacing 2m). The obstacle coverage rate is 30%. In the experiment, the coordinates of the starting point are (2m, 2m) and the coordinates of the target point are (18m, 18m). Changing the environmental information to verify that the optimization effect of the improved algorithm is obvious.

### 4.1 Comparative Analysis of A* Algorithm and Improved A* Algorithm Simulation Experiment

An improved A* algorithm and a traditional A* algorithm are planned in the initialization maps shown in Figures 3 and 4. The dot indicates the robot starting point and the diamond indicates the robot end point.

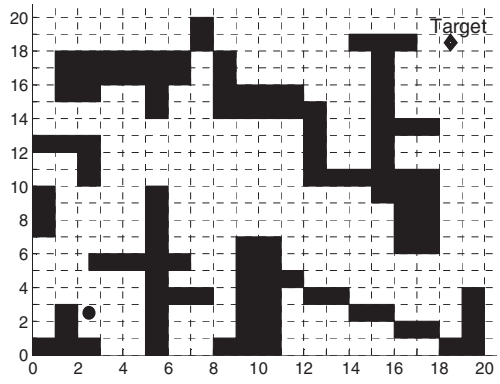



Fig.3 Initialization indoor map



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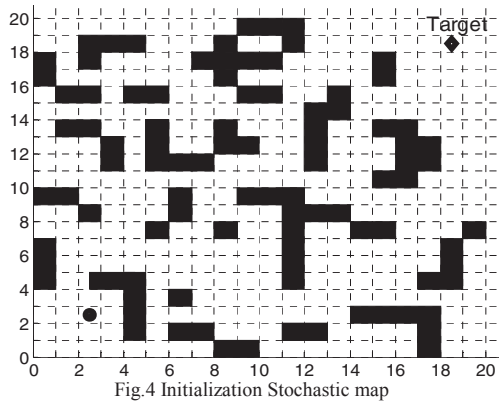


Fig.4 Initialization Stochastic map

Figure 3 for the general indoor environment. The results of the A* algorithm and the improved A* algorithm global path planning are shown in Figure 5. Figure 4 is a randomly generated environment. The results of the A* algorithm and the improved A* algorithm global path planning are shown in Figure 6

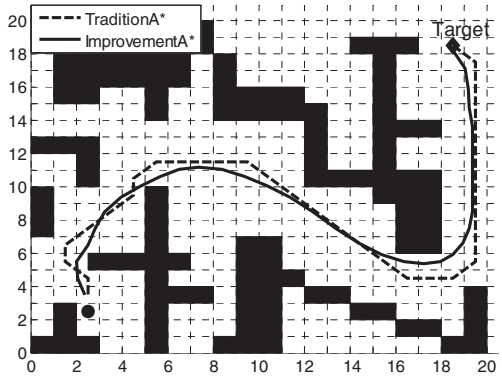


Fig.5 traditional A* and improved A* of indoor map

The solid line in Figure 5 is the path of the improved A* algorithm, and the dotted line is the path of the traditional A* algorithm. Compare the results of the connection between the start and end points of the two algorithms. It can be seen that the path to improve A* is smoother than the path of the traditional A*. And the turning angle of the inflection point in the path is optimized.

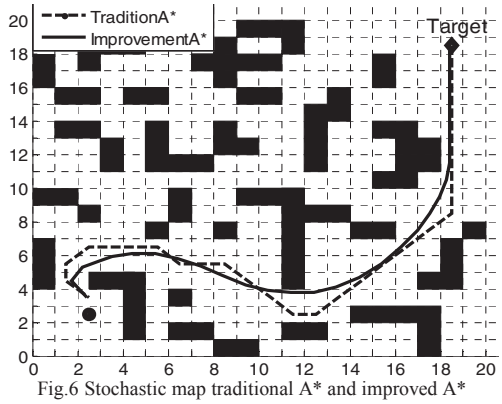


Fig.6 Stochastic map traditional A* and improved A*

The solid line in Figure 5 is the path of the improved A* algorithm, and the dotted line is the path of the traditional A* algorithm. The path planning of the random map in Figure 6 again shows that the improved A* is better. The path is smooth, the distance is shorter, and the turning point of the inflection point is better. Table 1 is the average value of 10 plans for the traditional A* algorithm and the improved A*.

**Table 1** Comparison of traditional and improved A*

| | Average time | Average length | Path node | Smoothing |
|---|---|---|---|---|
| Traditional A* | 19.22 | 37.62 | 34 | NO |
| Improved A* | 17.13 | 33.75 | 25 | YES |
| optimization | -10% | -10% | -26% | smooth |

It is seen from Table 1 that the improved A* algorithm is improved in many ways. The path length is reduced by 15%, the time spent is reduced by 8%, and the number of nodes is reduced by 36%. The traditional A* algorithm is optimized in terms of time, distance, number of nodes, and so on.

### 4.2 Analysis of Dynamic Obstacle Avoidance Effect after Introducing Morphin Algorithm

Firstly the robot travels according to the global path planned by the improved A* algorithm. It is assumed that there are three dynamic obstacle balls A, B and C in the grid diagram during robot driving. The experimental parameters are as follows: The average speed of the ball is 0.23m/s, and the average speed of the car is 2.2m/s. The position of the balls are A (6m, 10m); B (11m, 8m); C (15m, 13m). The obstacle and starting point position in the map are shown in Figure 7. The RRT algorithm searches for 10 nodes per step and the maximum number of searches is 1000. The RRT search tree algorithm performs comparison experiments under the same map as shown in Figure 10. It compare with the time, path, and accuracy of dynamic algorithms.

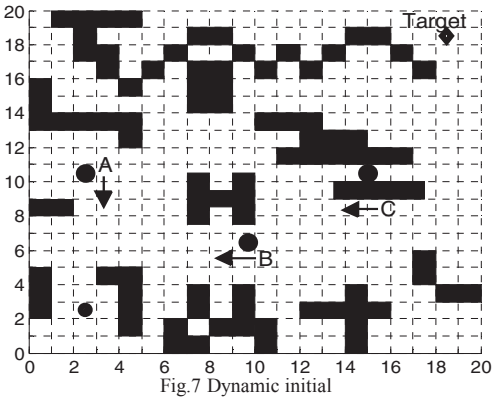


Fig.7 Dynamic initial

The Morphin algorithm predicts whether a robot collides with an obstacle. Then perform local path planning as shown in Figure 8. The spot line path in the figure is the original path, and the dotted line is the predicted arc generated by the Morphin algorithm. Because the movement direction of the small ball A does not interfere with the direction of the robot, the robot can avoid the small ball A according to the global path. It can be seen from the prediction that the N1 point of the broken line will collide with the small ball B. The car selects the path that predicts the smallest arc f to perform local obstacle avoidance. It



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avoids obstacles and returns to the global path to continue driving. The moving ball C causes the robot to fail in the passage between the obstacles. There are two scenarios for sports ball C：In the first case, the small ball C speed is greater than 0.4 m/s, which comes out of the channel in a short time and does not block the movement of the robot. As shown in Figure 8. The robot waits for the ball C to move out at point N2. Then drive to the target point according to the planned global path.

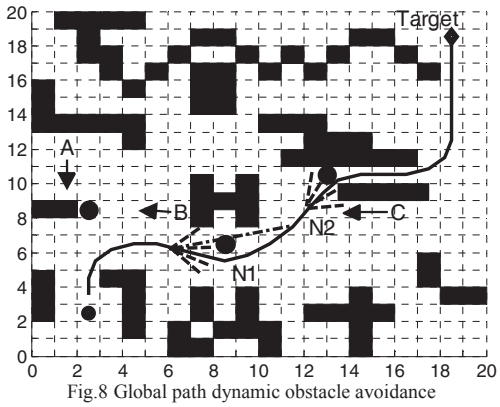


Fig.8 Global path dynamic obstacle avoidance

In the second case, the small ball C speed is less than 0.1 m/s, and stops in the channel after the movement. The robot cannot travel according to the planned global path. The passage between the obstacles is blocked as shown in Figure 9. At this point, the robot recalls the improved A* algorithm to re-plan the global path. It takes a new planned route to the target point.

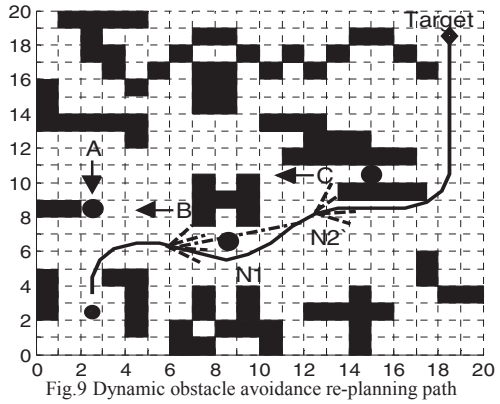


Fig.9 Dynamic obstacle avoidance re-planning path

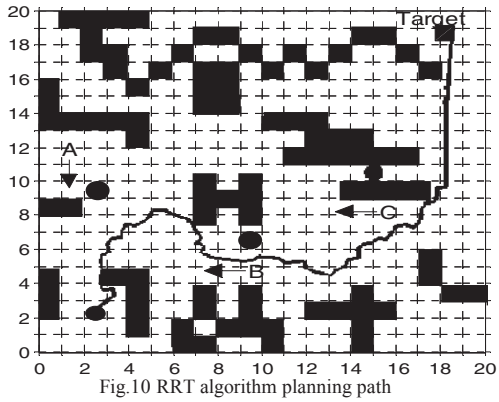


Fig.10 RRT algorithm planning path

Figure 9 is the same as the static and dynamic obstacles in Figure 10. Dynamic algorithm searches for shorter paths and better results. Because the RRT algorithm itself generates a random tree to find the path. It lacks repeatability and the path found is not optimal. Table 2 is the data comparison of A* algorithm, improved A* algorithm, RRT algorithm and dynamic algorithm.

Table.2 Path planning algorithm comparison result

| | Average time | Average length | Path node | Smoothing |
|---|---|---|---|---|
| Traditional A* | 19.42 | 38.04 | 34 | NO |
| Improved A* | 17.06 | 33.12 | 25 | YES |
| RRT algorithm | 31.75 | 52.47 | 87 | NO |
| dynamic algorithm | 17.98 | 35.42 | 28 | YES |

Table 2 shows the average value of each path planning algorithm running 10 times separately. Although the time and path length of the dynamic algorithm are greater than the improved A* algorithm, the dynamic algorithm is superior to the traditional A* algorithm. It takes less time and has a shorter path to ensure that the robot reaches the target safely and efficiently.

## 5 CONCLUSION

In this paper, the traditional A* algorithm faces the collision of dynamic obstacles or the failure of path planning. A global dynamic path planning method combining improved A* algorithm and Morphin algorithm is proposed. Firstly, A key node selection strategy is designed to remove redundant nodes in the traditional A* algorithm planning path sequence, so as to improve the performance of path planning. Local obstacle avoidance can be realized based on Morphin algorithm. Through the fusion of the two algorithms, the robot obtains the best walking path and the time is shorter, which effectively improves the working efficiency and safety of the robot path planning. The simulation test proves that the robot selects the optimal path under multiple obstacles, In the presence of dynamic obstacles, a local path can be quickly planned to avoid obstacles.

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